

STAINLESS STEEL MANUFACTURING · STEEL MELTING SHOP

AOD Operations

Process Analysis, Standardisation and Business Requirements

Portfolio edition. This is a sanitised version of work carried out in an operations role at a large stainless steel producer. The employer is not named, and all process parameters, quantities, temperatures, gas ranges, equipment identifiers and timings shown are illustrative placeholders, not operating values. The analysis structure, business rules, requirements, controls and validation logic reflect the work as performed. Nothing commercially sensitive is reproduced.

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About This Document

This document consolidates three pieces of process and requirements work carried out in the AOD (Argon Oxygen Decarburisation) area of a Steel Melting Shop. Each part addresses a distinct operational problem, and each follows the same structure: business problem, objectives and scope, stakeholders, current-state analysis, future-state design, rules and controls, and the measures used to confirm the change held.

The AOD converter is the point at which carbon is removed from the melt and the final chemistry of the heat is set. Material additions, gas ratios and temperature windows differ by grade family, and the process runs continuously across three shifts. The recurring problem across all three pieces of work is the same one: operating knowledge sat with experienced individuals rather than in a controlled reference, so execution varied by shift and by operator.

The work was performed in an operations role. The documentation approach, meaning structured requirements, business rules, traceability and acceptance criteria, is the approach used in business analysis, applied here to a manufacturing process rather than a software system.

Document Control

Field	Detail
Prepared by	S. Satyam Singh
Role at the time	Associate Manager, AOD Operations
Business area	Steel Melting Shop, AOD Operations
Coverage	Grade-wise addition pattern standardisation; AOD auto batching requirements; 5S material allocation
Methods applied	Process mapping (As-Is / To-Be), requirements definition, business rules, DMAIC, RACI, control planning, validation scenarios, traceability
Grade families	200 series, 300 series, 400 series
Version	2.0 (portfolio edition)

Contents

Part	Subject	Focus
A	Grade-Wise AOD Addition Pattern	Process mapping and standardisation of material addition sequence, temperature control and gas switch-over across three grade families
B	AOD Auto Batching	Business and functional requirements, bunker and weighing-hopper logic, exception handling and validation scenarios for automated material batching
C	5S Material Allocation at AOD	Current-state analysis, gap analysis and requirements for material location control, receipt handover and space utilisation

Terms Used

Term	Meaning
AOD	Argon Oxygen Decarburisation, the converter process used to refine stainless steel
Main blow / DB1 / DB2	Primary oxygen blowing stage, followed by second and third blow stages
Zero batch	Initial flux charge added before pouring
Dololime / Lime / Flux	Slag-forming additions used to control chemistry and protect refractory
Ferroalloys	Chromium, nickel and manganese bearing additions used to achieve target chemistry
MHS	Material Handling System: conveyors, bunkers, feeders and chutes serving the converter
CH1 / CH2	Charging hoppers feeding material into the converter
Weighing hopper	Intermediate hopper that weighs material before transfer to the charging chute
Reduction	Post-blow stage where chromium is recovered from slag

PART B

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AOD Auto Batching

Business and functional requirements, operating rules, exception handling and validation scenarios for automated material batching during the main blow window.

B1. Background

AOD operations require metallic and flux additions during the main blow period. Automated batching depends on accurate material selection, correct bunker configuration, appropriate feed and discharge rates, and clear trigger points tied to the oxygen phase.

For the sequence to run without interruption, material shifting, charging hopper movement, feeder condition, belt status, flame protection gates and auto-mode readiness must all be controlled before and during the process.

B2. Business Problem

The operational challenge was to control auto batching steps repeatably so that material additions complete within the expected main blow window. Without a structured process view, teams faced ambiguity around bunker priority, weighing hopper usage, colour status interpretation, feeder trips, charging hopper readiness and step table reconciliation.

Problem area	Business impact
Material addition timing	Delay in completing main blow additions and risk of process variation
Bunker and weighing hopper logic	Incorrect priority or simultaneous charging created operational confusion
Status visibility	Operators required clear colour-coded status to understand the material movement stage
Charging hopper control	Material must be fully discharged and tare weight zero before the next shift of material
Safety and exception handling	Feeder trip, converter tilting and power failure required clear control actions
Evidence and reconciliation	Step table output required tallying against the discharge table after addition

B3. Objectives

Objective
Document a standard process for AOD auto batching execution
Define business rules for bunker priority, material shifting, weighing hopper use and charging hopper readiness
Improve visibility by documenting colour-coded process status rules
Support completion of metallic addition and major flux addition within the main blow period
Reduce operational ambiguity between AOD, MHS, Electrical, Mechanical and shift teams
Create validation scenarios usable for process checks, training and system acceptance

B4. Scope

In scope	Out of scope
Auto batching process flow for all three grade families	Steel grade chemistry design and metallurgical formula ownership
Feed rate, discharge rate and material quantity requirements	Procurement and supplier commercial decisions
Bunker, weighing hopper and charging hopper operating rules	PLC coding and automation software development
Colour-coded status interpretation	Hardware design of conveyors, gates and feeders
Exceptions including feeder trip, converter tilting and power failure	Final approval authority for plant operating standards
Validation scenarios for process acceptance	Financial business case beyond process improvement logic

B5. Stakeholders

Stakeholder	Role in process	Requirement focus
Shift-In-Charge	Owns shift-level execution and decision making	Clear steps, exceptions, tally checks and safety controls
AOD Operator	Runs material addition steps and monitors the converter	Oxygen trigger points, charging hopper readiness, discharge status
MHS Team	Responsible for material handling equipment and feeders	Feeder health, belt status, bunker movement, chute availability
Electrical Team	Supports electrical readiness and flame protection gates	Auto mode, gate health, power-related exception response
Mechanical Team	Supports converter positioning and mechanical issues	Cooling position, back pressure checks, emergency handling
Production Management	Tracks productivity and process performance	Main blow timing, process discipline, shift compliance
Quality / Process Team	Checks grade requirements and output stability	Controlled chemistry and consistency of material addition

B6. Step Table Structure

Structure as documented. All values below are illustrative placeholders and do not represent operating parameters.

Step	Bunker	Trigger	Material	Qty	Feed rate	Discharge rate	Shift time	Remarks
S0	B-a	Prior heat, third blow	Slag former	Q1	n/a	n/a	0	Shift during previous heat
S1	B-b	Early oxygen band	Slag former	Q1	n/a	n/a	0	Shift when pouring starts; hold reserve on charging chute
S2	B-c & B-d	Oxygen band 2	Chromium alloy	Q2	F1	D1	T1	Shift as soon as confirmed
S3	B-e & B-f	Oxygen band 3	Metallic addition	Q3	F2	D2	T2	
S4	B-g & B-h	Oxygen band 4	Manganese alloy	Q4	F1	D1	T3	
S5	B-i	Oxygen band 5	Flux	Q5	F3	D3	T4	
Total	Expected total process time reference, compared against the main blow window							

A separate step table was maintained for each of the three grade families. The 300 series table carries additional nickel-bearing steps and therefore a longer expected total; the 400 series carries the shortest.

B7. As-Is and To-Be Process View

Area	As-Is observation	To-Be requirement
Process reference	Operational rules held across notes, step tables and team knowledge	A single structured process document for execution and training
Material movement	Material shifting depended on correct understanding of bunker and weighing hopper rules	Defined priority rules and conditions for simultaneous or sequential bunker operation
Status understanding	Operators interpreted colour-coded states from experience	Each status colour and its meaning documented
Safety controls	Critical exceptions handled through SOP and shift experience	Exception rules included in the process control section
Validation	Step table and discharge table comparison required after addition	Reconciliation defined as a formal post-addition control

B8. Functional Requirements

ID	Area	Requirement	Priority
FR-01	Series selection	The process shall support auto batching step tables for each grade family	High
FR-02	Material setup	The process shall define material name, quantity, bunker, feed rate and discharge rate for each step	High
FR-03	Oxygen trigger	Each material addition step shall be linked to a defined oxygen trigger or process event	High
FR-04	Colour status	The process shall follow colour-coded status for filling, shifting, ready or discharging, and completed states	High
FR-05	Bunker priority	Where multiple bunkers use the same weighing hopper in one step, a fixed left-to-right priority shall be followed	High
FR-06	Parallel bunker use	Where bunkers use different weighing hoppers in one step, simultaneous start shall be allowed	Medium
FR-07	Charging hopper control	Material shall shift only after the previous material is fully discharged and tare weight is zero	High
FR-08	Belt status	Material shall shift only when belt status is green; red status shall block shifting	High
FR-09	Auto mode	Equipment required for auto batching shall remain in auto mode unless exception handling is active	High
FR-10	Post-addition check	After addition, the step table shall be tallied against the discharge table	High
FR-11	Tilting control	During converter tilting in main blow, both charging hoppers shall be discharge-stopped before tilting	High
FR-12	Emergency handling	During power failure or critical interruption, additions shall stop and emergency checks shall be initiated	High

B9. Business Rules

Rule ID	Business rule
BR-B01	Metallic addition and major flux addition must be completed within the main blow planning window
BR-B02	A dedicated bunker is allocated to the primary slag former as part of the auto batching configuration
BR-B03	Bunkers not required for a specific grade family must be marked out of service
BR-B04	Status colour 1 means weighing hopper filling is in progress or not yet complete
BR-B05	Status colour 2 means material is shifting to the charging chute
BR-B06	Status colour 3 means all material has shifted to a charging hopper
BR-B07	Status colour 4 means material is ready to discharge or is discharging
BR-B08	Status colour 5 means the step is complete
BR-B09	For the same weighing hopper, bunker priority follows a fixed sequence and must not start simultaneously
BR-B10	For different weighing hoppers, bunkers may start simultaneously within the same step
BR-B11	The feeder must not trip during charging or discharging; MHS owns feeder readiness
BR-B12	The outlet flame protection gate must be healthy and inlet gates manually open where required
BR-B13	After zero batch is shifted to a charging hopper, the empty source must be marked out of service
BR-B14	On main blow power failure, the lance emergency control must be operated so the lance returns to rest position

B10. Data Requirements

Data field	Purpose
Series	Identifies the applicable grade family step table
Step	Defines sequence number for the batching step
Bunker	Identifies source bunker for material
Oxygen trigger	Defines process event or oxygen trigger for addition
Material name	Defines material to be added
Material quantity	Defines target material quantity
Feed rate	Defines material feeding rate
Discharge rate	Defines belt discharge rate
Time to shift chute	Calculates expected shift time
Remark	Captures execution condition or operational note

B11. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Safety	The process must prevent unsafe material shifting during converter tilting, power failure or equipment abnormality
NFR-02	Usability	Status colours and step tables must be readily interpretable by operators and Shift-In-Charges
NFR-03	Traceability	Step table and discharge table must be reconcilable after addition
NFR-04	Reliability	Feeder, gates and auto-mode readiness must be verified to reduce process interruption
NFR-05	Maintainability	Business rules must be straightforward to update when grade, bunker or material configuration changes

B12. Exception Handling and Control Plan

Exception	Immediate control	Owner	Evidence
Feeder trip during charging or discharging	Stop, check feeder health and resolve before continuing material movement	MHS Team	Equipment status and shift log
Belt shows red status	Do not shift material until status is green	AOD Operator / MHS	Status indication
Material not fully discharged	Do not shift next material until discharge is complete and tare weight is zero	AOD Operator	Tare weight check
Converter tilting during main blow	Both charging hoppers must be discharge-stopped before tilting	Shift-In-Charge / AOD Operator	Operator confirmation
Power failure during blowing	Stop all additions, confirm emergency gas valve, call Electrical and Mechanical, move converter to cooling position as required	Shift-In-Charge / Electrical / Mechanical	Power failure response log
Main blow power failure	Operate the lance emergency control so the lance returns to rest position	AOD Operator	Emergency action confirmation

B13. Validation Scenarios

ID	Precondition	Action	Expected result
V-01	A grade family is selected	Run that family's step table	Sequence follows S0 to the final step and total time falls within the expected reference
V-02	300 series selected	Run the 300 series step table	Nickel, chromium, metallic, manganese and flux steps follow the defined sequence
V-03	Same weighing hopper used by multiple bunkers	Trigger all three bunkers in the same step	The first starts, then the others in sequence; simultaneous start is blocked
V-04	Different weighing hoppers used	Trigger all three bunkers in the same step	All permitted bunkers may start simultaneously
V-05	Belt status red	Attempt material shift	Material shift is not allowed
V-06	A charging hopper holds a material balance	Attempt next material shift	Process waits until material is fully discharged and tare weight is zero
V-07	Converter tilting during main blow	Initiate tilting control	Both charging hoppers are discharge-stopped before tilting
V-08	Power failure during blowing	Apply emergency process	Additions stop, emergency gas valve is checked, Electrical and Mechanical are called
V-09	Addition complete	Compare step table with discharge table	Material quantities and steps reconcile

B14. Requirements Traceability

Requirement	Linked rule or control	Validation	Business value
FR-01	Series-wise step tables	V-01, V-02	Supports grade-wise process control
FR-02	Material setup table	V-01, V-02	Ensures quantity and material detail are controlled
FR-04	Colour status rules	V-05	Improves operator visibility
FR-05	Same weighing hopper priority rule	V-03	Controls sequential bunker operation
FR-06	Different weighing hopper simultaneous rule	V-04	Controls parallel bunker operation
FR-07	Tare weight and discharge completion	V-06	Prevents incorrect material movement
FR-10	Step table against discharge table tally	V-09	Creates post-addition evidence
FR-11	Tilting control	V-07	Controls safety risk
FR-12	Emergency handling	V-08	Controls power failure response

B15. RACI Matrix

Activity	Shift-In-Charge	AOD Operator	MHS	Electrical	Mechanical	Production
Confirm grade and series step table	A	R	C	I	I	C
Verify material and bunker readiness	A	R	R	I	I	C
Check auto mode and flame gates	A	R	C	R	C	I
Run material shifting and discharge steps	A	R	R	I	I	I
Handle feeder trip	A	C	R	C	C	I
Handle power failure	A	R	C	R	R	I
Tally step table and discharge table	A	R	C	I	I	C

R = Responsible · A = Accountable · C = Consulted · I = Informed